

Creating a Maven Project with Guava and Commons-IO

Maven Project with Guava and Commons-IO

Dependencies Create a Maven Project Using Terminal by following command:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
samiksha@samiksha:~/report$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/samiksha/report
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/report/MyMavenGuavaApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.360 s
[INFO] Finished at: 2026-03-17T19:29:09+05:30
[INFO] -----
samiksha@samiksha:~/report$
```

Now Navigate to the Project Directory using **cd MyMavenGuavaApp**

```
samiksha@samiksha:~/report$ cd MyMavenGuavaApp
samiksha@samiksha:~/report/MyMavenGuavaApp$
```

Edit the pom.xml File and add the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenGuavaApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenGuavaApp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
```

</dependency>

<!-- https://mvnrepository.com/artifact/com.google.guava/guava -->

<dependency>

<groupId>com.google.guava</groupId>

<artifactId>guava</artifactId>

<version>33.4.0-jre</version>

</dependency>

<!-- https://mvnrepository.com/artifact/commons-io/commons-io -->

<dependency>

<groupId>commons-io</groupId>

<artifactId>commons-io</artifactId>

<version>2.18.0</version>

</dependency>

</dependencies>

<!-- Build: Configuring plugins and build settings -->

<build>

<plugins>

<!-- Example: Maven Compiler Plugin to compile Java code -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.8.1</version>

<configuration>

<source>1.7</source>

<target>1.7</target>

</configuration>

</plugin>

<!-- Example: Maven Surefire Plugin to run tests -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-surefire-plugin</artifactId>

<version>2.22.2</version>

</plugin>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-jar-plugin</artifactId>

<version>3.1.0</version>

<configuration>

<archive>

```

        <manifestEntries>
            <Main-Class>com.example.App</Main-Class>
        </manifestEntries>
    </archive>
</configuration>
</plugin>

</plugins>
</build>
</project>

```

Modify App.java

Got to App.java using **cd src/main/java/com/example/** and **gedit App.java** to type the file with following code:

```
package com.example;
```

```

/**
 * Hello world!
 *
 */
// Import FileUtils class from Apache Commons IO library
// This is used to perform file operations like copying files
import org.apache.commons.io.FileUtils;

// Import File class to represent file and directory pathnames
import java.io.File;

// Import IOException to handle input-output errors
import java.io.IOException;

// Import ImmutableList from Google Guava library
// ImmutableList creates a list that cannot be modified after creation
import com.google.common.collect.ImmutableList;

// Define the main class
public class App
{

```

```

// Main method - program execution starts from here
public static void main( String[] args )
{
    // Create an immutable list of strings (fruits)
    // This list cannot be changed (no add/remove allowed)
    ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");

    // Print the list of fruits to the console
    System.out.println(fruits);

    // Create a File object representing the source file
    File sourceFile = new File("source.txt");

    // Create a File object representing the destination file
    File destFile = new File("destination.txt");

    try {
        // Copy the contents of source.txt to destination.txt
        // Using Apache Commons IO FileUtils class
        FileUtils.copyFile(sourceFile, destFile);

        // Print success message if file is copied
        System.out.println("File copied successfully!");
    }
    catch (IOException e) {
        // If any error occurs during file copying,
        // this block will handle the exception
        // Print error message
        System.err.println("Error occurred while copying file: " + e.getMessage());
    }
}

```

Build the Project using the command **mvn clean install** command

```
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.2.0/maven-archiver-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.2.0/maven-archiver-3.2.0.pom (4.3 kB at 144 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.2.0/maven-shared-utils-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.2.0/maven-shared-utils-3.2.0.pom (4.9 kB at 189 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.5/plexus-archiver-3.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.5/plexus-archiver-3.5.pom (4.8 kB at 185 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/5.0/plexus-5.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/5.0/plexus-5.0.pom (21 kB at 790 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.0.0/plexus-io-3.0.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.0.0/plexus-io-3.0.0.pom (4.5 kB at 175 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.14/commons-compress-1.14.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.14/commons-compress-1.14.pom (13 kB at 505 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/42/commons-parent-42.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/42/commons-parent-42.pom (68 kB at 2.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.6/xz-1.6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.6/xz-1.6.pom (1.9 kB at 73 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.1.0/plexus-utils-3.1.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.1.0/plexus-utils-3.1.0.pom (4.7 kB at 155 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.2.0/maven-archiver-3.2.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.2.0/maven-archiver-3.2.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.5/plexus-archiver-3.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.5/plexus-archiver-3.5.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.0.0/plexus-io-3.0.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.0.0/plexus-io-3.0.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.14/commons-compress-1.14.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.14/commons-compress-1.14.jar (24 kB at 593 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.6/xz-1.6.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.6/xz-1.6.jar (74 kB at 1.8 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.1.0/plexus-utils-3.1.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.1.0/plexus-utils-3.1.0.jar (187 kB at 3.9 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.5/plexus-archiver-3.5.jar (187 kB at 3.9 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.6/xz-1.6.jar (103 kB at 1.4 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.1.0/plexus-utils-3.1.0.jar (262 kB at 3.2 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.14/commons-compress-1.14.jar (530 kB at 5.4 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.2.0/maven-shared-utils-3.2.0.jar (165 kB at 1.6 MB/s)
[INFO] Building jar: /home/unified/Devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/unified/Devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /home/unified/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/unified/Devops/MyMavenGuavaApp/pom.xml to /home/unified/.m2/repository/com/example/MyMavenGuavaApp/1.0-SNAPSHOT/MyMavenGuavaApp-1.0-SNAPSHOT.pom
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.093 s
[INFO] Finished at: 2025-05-28T21:31:52+05:30
[INFO]
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

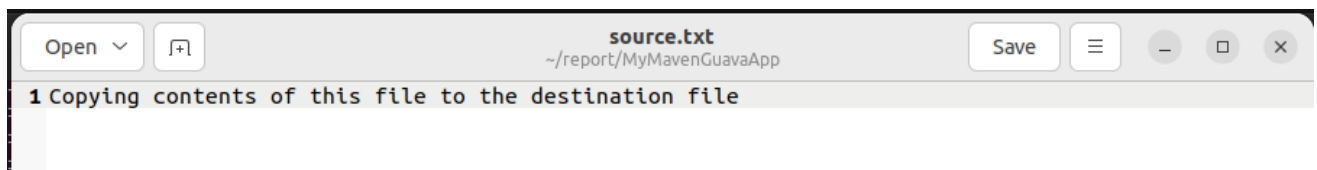
If java-jar command give an error which is most probable, use the following command **mvn exec:java -Dexec.mainClass="com.example.App"**

```
samiksha@samiksha:~/report/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
Error occurred while copying file: Source 'source.txt' does not exist
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.559 s
[INFO] Finished at: 2026-03-17T20:28:33+05:30
[INFO] -----
```

This command gives error as

Error while copying file:Source ‘source.txt’ does not exist

Now, Create a Sample File to copy the content to destination file



Now Run the command **mvn exec:java -Dexec.mainClass="com.example.App"** again

```
samiksha@samiksha:~/report/MyMavenGuavaApp$ gedit source.txt
samiksha@samiksha:~/report/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.326 s
[INFO] Finished at: 2026-03-17T20:31:05+05:30
[INFO] -----
samiksha@samiksha:~/report/MyMavenGuavaApp$
```

The output shows file copied successfully.

We can see newly created **destination.txt** using gedit destination.txt if you are still in Project's directory

```
samiksha@samiksha:~/report/MyMavenGuavaApp$ ls
destination.txt  pom.xml  source.txt  src  target
samiksha@samiksha:~/report/MyMavenGuavaApp$
```

File content is same as the content in source file.

Step 1: now open the terminal and type the command **cd MyMavenGuavaApp** to enter the folder and create a **Jenkinsfile** .Paste the code in the Jenkinsfile

```
pipeline {
    agent any // Use any available agent
    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git
                branch: 'master',
                url: 'https://github.com/Samiksha201104/MyMavenGuavaApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package' // Run Maven build
            }
        }
        stage('Test') {
```

```

steps {
    sh 'mvn test' // Run unit tests
}
}

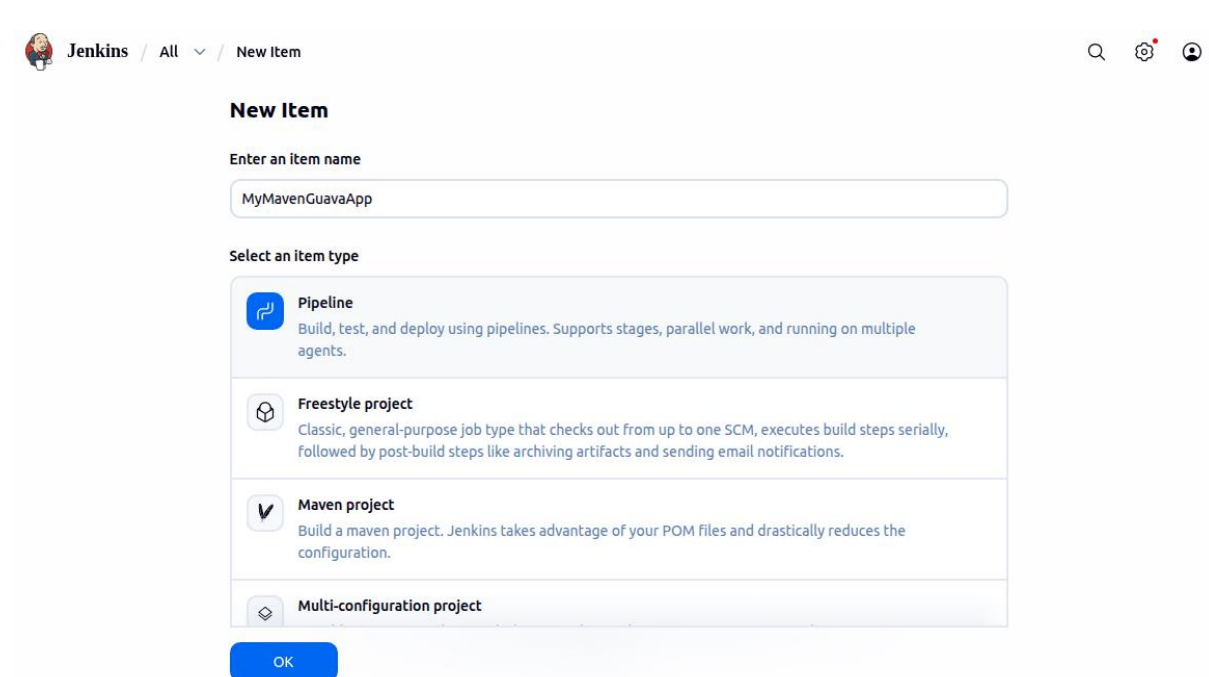
stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenGuavaApp-1.0-SNAPSHOT.jar'
    }
}
}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}

```

Then commit and push it to GitHub

Step 2: in the Jenkins Dashboard click on new item type the name of the item and select Pipeline



Jenkins / All / New Item

New Item

Enter an item name

MyMavenGuavaApp

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project**

OK

In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credentials

Configure

General Triggers Pipeline Advanced

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Samiksha201104/MyMavenGuavaApp.git

Please enter Git repository.

Credentials ?

GitHub Credentials + Add

Advanced

Save Apply

Step 3 : in the Dashboard click on the MyMavenGuavaApp item and select Build Now and then Full Stage view
Correct the errors and build again on jenkins till all the stages are successful

Jenkins / MyMavenGuavaApp / Full Stage View

MyMavenGuavaApp - Stage View

			Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Run Application	Declarative: Post Actions
Average stage times: (full run time: ~31s)			1s	269ms	2s	18s	958ms	566ms
#5	Apr 12 20:42	1 commit	1s	301ms	1s	21s	958ms	219ms
#4	Apr 12 20:40	1 commit	1s	224ms	1s	5s failed	236ms failed	283ms
#3	Apr 12 20:37	3 commits	1s	244ms	2s	27s	2s failed	1s
#2	Apr 12 20:33	4 commits	1s	307ms	3s	18s failed	502ms failed	537ms

